Interview Handbook

100+ questions by category and difficulty, each with what the interviewer tests, a structure, a model answer and common mistakes.



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Built from the 55-lecture course study guide and extended to the 12-module Product Core curriculum.
Hypothetical examples are labelled; no real company results are invented.

PART

Interview Handbook

100+ questions by category and difficulty, each with what the interviewer tests, a structure, a model answer and common mistakes.

14 entries

How to use this interview handbook (scorecards and worksheets)

106 questions across 13 categories, tagged **Easy / Medium / Hard**. Each has: *what the interviewer is testing*, a *structured approach*, a *model answer*, *assumptions and trade-offs*, *common mistakes*, and where useful a *60-second version*. Model answers are examples of good reasoning, **not scripts** — interviewers detect memorised answers. All company scenarios are hypothetical unless stated.

From your course notes (Lectures 22, 30, 52, 53)

Introduce yourself with relevant evidence (current role and product impact first). Explain your product, users, data flows and your *actual* contribution; distinguish outcomes from tools (story points are not success). **Clarify which metric fell before building a root-cause analysis.** Prepare honest STAR stories; **do not misrepresent your job title, invent metrics or claim work you did not do** — describe course and personal projects as such. Lecture 44 (Interview Prep 4) had no recording and is not summarised.

A universal answer structure



CIRCLES-style structure for product questions (adapt, don't recite). CIRCLES (popularised by Lewis Lin) is one of several mnemonics. Spend most time on users, needs and trade-offs.

Mock interview scorecard

Dimension	1 · Weak	3 · Solid	5 · Excellent	Score
Structure & clarity	Rambling, no framing	Clear structure, signposting	Crisp, adaptive, summarises well	
Clarifying & assumptions	Jumps in	Asks key questions	Prioritises the questions that change the answer; states assumptions	
User empathy	Generic users	Specific segment and needs	Deep insight into jobs, pains, contexts	
Problem solving & creativity	One obvious idea	Several sensible options	Novel yet feasible; considers non-tech fixes	
Prioritisation & trade-offs	No trade-offs	Explicit criteria	Quantified, with risks and sequencing	
Metrics & analytics	Vanity metrics	Primary + guardrail	Precise definitions, causal testing, segment thinking	
Technical judgement	Hand-wavy	Correct basics	Anticipates failure modes, cost, scale	
Communication & presence	Unclear	Clear	Engaging, concise, handles pushback	
Integrity	Overclaims	Honest	Transparent about uncertainty and own contribution	

Bar guidance (rough): average ≥ 3.5 with no 1s for a PM offer at most levels; senior roles need 4+ on strategy, trade-offs and leadership.

Practice worksheet (copy for each question)

Prompt	Your notes
Question & category	
What is being tested?	
Clarifying questions (top 3)	
Assumptions I'll state	
Users/segment chosen and why	
Key needs/pains	
Options (≥ 3)	
Choice + trade-offs	
Metrics (primary, guardrails)	
Risks & next step	
60-second summary	
Self-score (scorecard) and one thing to improve	

Practice plan

Week 1: product sense/design (2/day). Week 2: analytics, RCA, estimation. Week 3: strategy, prioritisation, technical. Week 4: AI PM, behavioural, leadership + 3 full mocks with a partner using the scorecard. Record yourself; review against the rubric.

Product sense (8 questions)

PS1 · Easy — What is your favourite product and why?

What it tests: product taste, user empathy, ability to articulate value and trade-offs.

Approach: product + user + job → why it works (2–3 design/business reasons) → one meaningful weakness with evidence → improvement + metric.

Model answer: “Google Maps for commuting. My job is ‘get to work on time with least stress.’ It works because (1) live traffic turns uncertainty into an ETA I trust, (2) it re-routes proactively, (3) it combines modes. A weakness for me: on crowded metro lines it doesn't tell me how full trains are, so I can't choose a less crowded one. I'd test showing crowding levels from anonymised, aggregated data, measuring route-choice changes and satisfaction, with privacy as a guardrail.”

Assumptions & trade-offs: crowding data availability and accuracy; privacy vs usefulness; added screen complexity.

Common mistakes: listing features; choosing a product you can't critique; no user or metric.

60-second answer: job → two reasons it works → one improvement → how you'd measure it.

PS2 · Easy — What makes a product great?

What it tests: principles over buzzwords; whether you connect user value to business sustainability.

Approach: define “great” (value, usability, viability, trust) → evidence signals → one example.

Model answer: “A great product reliably helps a specific user make progress on an important job, with as little effort and risk as possible, in a way the business can sustain. Signals: users return without being pushed (retention curves flatten), they recommend it, and the economics improve with scale. Example: a payments flow that just works at checkout — invisible reliability is a feature.”

Assumptions & trade-offs: “great” differs by segment; delight vs simplicity; growth vs trust.

Common mistakes: “beautiful design” alone; ignoring economics or reliability.

60-second answer: the definition above plus one example and one signal.

PS3 · Medium — Critique the onboarding of an app you used recently.

What it tests: observation, heuristics, prioritisation.

Approach: goal of onboarding (time-to-value) → walk the steps → friction tied to heuristics → top fix → metric.

Model answer: "A budgeting app asked for bank linking, notifications and goals before showing anything useful. Heuristic: visibility and user control — I had no idea what value I'd get. Biggest fix: show a sample dashboard immediately and ask for bank linking in context when I tap 'see my spending'. Metric: activation (first budget created within 3 days); guardrails: permission grant rate, privacy complaints."

Assumptions & trade-offs: linking later may reduce data completeness; fewer upfront questions vs personalisation.

Common mistakes: critiquing visuals only; no prioritised fix; no measurement.

60-second answer: the job of onboarding → biggest friction → one fix → activation metric.

PS4 · Medium — Which product do you think is badly designed, and how would you fix it?

What it tests: constructive critique and empathy for constraints.

Approach: acknowledge constraints → specific user problem → evidence → fix → trade-off.

Model answer: "Many parking-payment apps: you must register, add a card and find a zone code while standing in the street. Constraint: municipalities require identification and payment records. Fix: a QR on the meter that opens a lightweight web flow with wallet payment, with optional account creation later. Trade-off: less data capture; mitigate with optional emailed receipts."

Assumptions & trade-offs: municipal rules allow guest payment; enforcement systems can read a plate-based session.

Common mistakes: mocking the product; ignoring why it is that way.

60-second answer: constraint → user pain → QR guest flow → trade-off.

PS5 · Medium — How would you improve YouTube Shorts?

What it tests: narrowing a broad prompt; multi-sided thinking (viewers, creators, advertisers); wellbeing guardrails.

Approach: clarify goal (engagement? creator retention?) → pick a segment → pain → solutions → metric + guardrails.

Model answer: "I'd focus on new viewers whose first Shorts feel irrelevant — they leave before personalisation kicks in. Ideas: a lightweight topic picker after the first few swipes; a 'not interested' that visibly teaches the feed; separation of short vs long results in search. I'd A/B the topic picker; primary: day-7 return rate of new viewers; guardrails: session-satisfaction surveys and time-spent spikes among younger users."

Assumptions & trade-offs: new-viewer churn is a real problem (verify with data); a picker adds friction.

Common mistakes: optimising only watch time; proposing features for every stakeholder at once. *(Course link: Lecture 11 — ideas were participants' hypotheses.)*

60-second answer: segment → cold-start pain → topic picker test → day-7 return + wellbeing guardrail.

PS6 · Medium — Should Netflix let users stop a title from leaving the catalogue?

What it tests: feasibility and business constraints (content rights) alongside user empathy.

Approach: validate the need → identify constraints → reframe to the underlying job → alternatives → metric.

Model answer: "The need is real — losing a show mid-watch is frustrating — but licences expire contractually, so a 'keep' button is usually infeasible. Better: warn earlier ('leaving in 14 days'), a 'finish before it leaves' row, and use the count of mid-watch members as a signal in licence negotiations."

Assumptions & trade-offs: licence terms vary; early warnings could increase churn intent if users dislike departures.

Common mistakes: proposing something contractually impossible without acknowledging it. *(Course link: Lecture 4.)*

60-second answer: need → constraint → alternatives → metric (completion before departure).

PS7 · Hard — What product would you build for elderly users to manage medicines?

What it tests: empathy, accessibility, safety, multi-stakeholder design.

Approach: users (elderly person, caregiver, pharmacist) → jobs → constraints (vision, dexterity, tech comfort) → solution → safety → metrics.

Model answer: "Start with a caregiver-set schedule and a very simple 'taken' confirmation (large button, voice option); missed-dose alerts to a caregiver after a grace period; pharmacy refill reminders. Avoid medical advice; partner with pharmacists for dose data. Metrics: doses confirmed on time; caregiver alerts resolved. Guardrails: false alarms, privacy consent."

Assumptions & trade-offs: a caregiver exists and consents; autonomy vs oversight; device choice (phone vs smart dispenser).

Common mistakes: complex app UI; giving medical advice; ignoring consent.

60-second answer: caregiver schedule + one-tap confirmation + alerts; adherence metric; safety guardrails.

PS8 · Hard — What is a product you think will fail, and why?

What it tests: judgement under uncertainty, humility, use of criteria.

Approach: state criteria (job, alternatives, switching costs, economics) → apply → what would change your mind.

Model answer: "I'd question standalone AI gadgets that duplicate phone capabilities: the job (quick answers) is already served by phones; the device adds cost and charging burden; switching costs favour phones. It could succeed only if it serves a job phones do badly — hands-free contexts or accessibility. I'd watch day-30 retention and daily use versus phone usage."

Assumptions & trade-offs: future capabilities could change the picture; niche success is still success.

Common mistakes: snark; no criteria; certainty without evidence.

60-second answer: criteria → why it struggles → when it could win → signal to watch.

Product design (9 questions)

PD1 · Medium — Design a search engine for children under 14.

What it tests: user segmentation (child vs parent), safety, trade-offs.

Approach: clarify age bands and purpose (school vs fun) → users (child, parent, teacher) → needs → design → safety → metrics.

Model answer: "Focus on 8–12-year-olds doing homework. Needs: readable results, trustworthy sources, no unsuitable content, parents' reassurance. Design: curated educational sources first, reading-level summaries, image search with strict filtering, 'ask a grown-up' prompts for sensitive topics, a parent dashboard with privacy-preserving summaries. Filtering is imperfect — don't promise complete safety; add a report button; no personalised ads. Metrics: task success (found an answer); parent trust score; guardrail: unsafe-content reports."

Assumptions & trade-offs: curation narrows coverage; children's privacy rules limit data use; engagement must serve learning, not screen time.

Common mistakes: one persona only; promising 100% safety. *(Course link: Lecture 8.)*

60-second answer: 8–12 homework segment → curated, readable, safe results → parent controls → task success + safety guardrail.

PD2 · Medium — Help a professional regain control of an overflowing inbox after a holiday.

What it tests: framing the job; AI risk management.

Approach: define the job → triage needs → design → AI risks and controls → metrics.

Model answer: "Job: find what's urgent and relevant without reading everything. Design: a 'Catch-up' view grouped by people I work with, deadlines and mentions; AI thread summaries with links to originals; bulk actions with preview and undo. Risks: misclassifying an important email, wrongly auto-archiving, exposing confidential data to AI — keep human review for bulk actions, test accuracy, respect permissions. Metric: time to clear important items; guardrail: 'missed important email' reports."

Assumptions & trade-offs: summaries may miss nuance; automation vs control.

Common mistakes: fully automatic archiving; ignoring data protection. *(Course link: Lecture 8.)*

60-second answer: catch-up view + summaries + safe bulk actions; time-to-clear metric with a missed-email guardrail.

PD3 · Medium — Design a feature to reduce food-delivery checkout abandonment.

What it tests: funnel thinking, UX states, measurement.

Approach: locate the drop (funnel) → cause (research) → design incl. failure states → experiment.

Model answer: "First confirm where users drop — say, the payment step. If research shows fee surprise, show an estimated total in the cart with 'est.' labels, refresh on address change, and an honest 'Estimate unavailable' fallback if the fee service fails. A/B test with checkout completion as primary; complaints and payment errors as guardrails." *(See Modules 5 and 7.)*

Assumptions & trade-offs: estimates may differ from final fees; showing fees earlier could reduce add-to-cart.

Common mistakes: redesigning everything; forgetting failure states.

60-second answer: funnel → fee surprise → early total with fallback → A/B with guardrails.

PD4 · Hard — Design a dating feature for a large social network.

What it tests: two-sided trust, safety, privacy, ethical monetisation.

Approach: goals and segments → core flows → safety/privacy → monetisation → metrics and guardrails.

Model answer: "Segment by intention (serious vs casual). Core: opt-in profile separate from the main profile, verified identity, mutual-interest reveal, events-based matching. Safety: consent, harassment reporting, location fuzzing, age verification, no exposure of private conversations. Monetisation: freemium features that don't compromise safety (no paying to message strangers). Metric: mutual matches that lead to conversations; guardrails: reports per 1,000 users, blocks."

Assumptions & trade-offs: separation from the main network vs leveraging its graph; compatibility scores may bias choice.

Common mistakes: engagement-only metrics; AI-invented targets. *(Course link: Lecture 41.)*

60-second answer: intentions → safe opt-in profiles + mutual reveal → ethical freemium → quality-match metric with safety guardrails.

PD5 · Medium — Design an ATM for the visually impaired.

What it tests: accessibility and inclusive design.

Approach: users and contexts → tasks (withdraw, balance) → barriers → design → testing.

Model answer: "Audio guidance via a headphone jack, a tactile keypad with raised markers, phone-based pre-staging (set the withdrawal in the app, then tap card/phone), screen blanking for privacy, consistent layouts and generous timeouts. Test with visually impaired users; measure task success and time; guardrail: fraud and shoulder-surfing incidents."

Assumptions & trade-offs: hardware retrofit cost; privacy vs audible prompts (use headphones).

Common mistakes: designing without involving visually impaired users; relying on colour or screens alone.

60-second answer: audio + tactile + phone pre-staging + privacy; test with real users.

PD6 · Medium — Design a parking app for a busy city.

What it tests: user segmentation, data constraints, operational realism.

Approach: segments → jobs → features → data/partners → metrics → risks.

Model answer: "Users: daily commuters vs occasional visitors. Jobs: find a spot quickly, pay easily, avoid fines. Features: live availability (sensors/partner garages), reserve and pay, extend remotely, reminders before expiry. Metrics: time to park, successful reservations, fine reduction (self-reported). Risks: inaccurate availability → show confidence levels."

Assumptions & trade-offs: data coverage limits; reserving spots may need operator agreements.

Common mistakes: assuming perfect real-time data; ignoring payment regulations.

60-second answer: commuter focus → find, pay, extend → time-to-park metric → data-confidence risk.

PD7 · Hard — Design a product to help small restaurants reduce food waste.

What it tests: B2B empathy, data feasibility, business value.

Approach: user (owner/chef) → job → simplest solution → data needs → metrics.

Model answer: "Job: prepare and order the right quantities. Start with a simple demand forecast from past sales + weather + local events, giving suggested prep quantities; later add a marketplace for end-of-day surplus. Low integration first (POS export/CSV). Metrics: waste % by weight, margin; guardrail: stock-outs."

Assumptions & trade-offs: restaurants record sales consistently; forecasts vs chef intuition (keep human control).

Common mistakes: heavy integrations first; ignoring stock-out risk.

60-second answer: forecast-based prep suggestions → waste % metric with stock-out guardrail.

PD8 · Medium — Design a feature to increase channel-creation quality in a team-chat tool.

What it tests: B2B UX, findability, experiment design at account level.

Approach: problem evidence → design at the moment of creation → metric → experiment unit.

Model answer: "Suggest similar existing channels when someone types a new channel name; offer naming templates and pre-filled purposes; improve directory search. Primary metric: duplicate-channel rate; secondary: channel-message share. Randomise by workspace to avoid contamination." *(See Case 7 and Project 3.)*

Assumptions & trade-offs: suggestions must respect private-channel permissions; small extra friction at creation.

Common mistakes: blocking creation; randomising at user level in a team product.

60-second answer: similar-channel suggestions + templates → duplicate rate → workspace-level A/B.

PD9 · Hard — Design an emergency feature for a ride-hailing app.

What it tests: safety-critical design and handling false positives.

Approach: stakeholders → scenarios → features → failure modes → compliance → metrics.

Model answer: "Users: rider, driver, emergency contacts, safety operations. Features: an SOS button that shares live location with emergency services/contacts per local rules, trip sharing, unusual-stop detection with a check-in prompt, and a safety centre. Design for false positives without delaying urgent calls; offline/poor-network fallback (SMS). Metrics: time to response, check-in resolution; guardrails: false alarms, privacy."

Assumptions & trade-offs: emergency-service integration varies by city; automation vs user control.

Common mistakes: treating it like a growth feature; ignoring local emergency integration.

60-second answer: SOS + sharing + anomaly check-ins; response-time metric; false-alarm and privacy guardrails.

Product strategy (8 questions)

ST1 · Medium — Should a food-delivery company launch grocery delivery?

What it tests: strategic fit, unit economics, capabilities.

Approach: goal → customer need → market size → right to win (assets) → economics → risks → test plan.

Model answer: "It's product development in an existing market (Ansoff). Assets: customer base, courier network, app. Need: fast top-ups for urban households. Economics differ: lower AOV and margin, inventory and dark stores. I'd pilot in two dense neighbourhoods with a limited assortment, measure repeat rate, contribution per order and cannibalisation of food orders, and scale only if contribution turns positive within ~6 months."

Assumptions & trade-offs: growth vs margin; focus vs expansion; operational complexity.

Common mistakes: "yes, big market"; ignoring unit economics and cannibalisation.

60-second answer: Ansoff framing → assets → different economics → limited pilot with contribution gate.

ST2 · Hard — How should a travel marketplace respond to a sudden collapse in travel (e.g., a pandemic)?

What it tests: strategy under crisis; demand and supply thinking.

Approach: what changed (demand, rules, trust) → segments still travelling → protect supply → product moves → metrics. *(Your course raised an Airbnb-during-COVID question in Lecture 10 without developing an answer; this is an independent, hypothetical approach — not a description of what any company did.)*

Model answer: "Focus on demand that still exists — nearby getaways and long stays for remote workers; adapt search to flexible destinations and monthly stays; strengthen trust with cleaning standards and flexible cancellation; support hosts with clear refund rules. Measure booking recovery by segment, host retention and refund disputes."

Assumptions & trade-offs: flexible cancellation increases host risk; cash preservation vs investment.

Common mistakes: only cost-cutting; ignoring the supply side.

60-second answer: shift to surviving segments, rebuild trust, protect hosts, measure recovery by segment.

ST3 · Medium — How would you enter a new country with a fintech product?

What it tests: market-entry structure, regulatory awareness.

Approach: PESTLE → segment → licence/partner model → localisation → pilot → economics.

Model answer: "Scan regulation, payments infrastructure and trust (PESTLE); choose a beachhead segment; partner with a licensed local bank first rather than seeking a licence; localise language, KYC documents and payment methods; pilot in one city. Key risks: regulatory timelines and new fraud patterns. Metrics: activation, fraud rate, CAC payback."

Assumptions & trade-offs: partner dependency vs speed; local compliance cost.

Common mistakes: copying the home-market product; ignoring licensing.

60-second answer: PESTLE → partner-led entry → localise → pilot with fraud and payback gates.

ST4 · Medium — A competitor launched a cheaper version of our product. What do you do?

What it tests: composure, segmentation, evidence before reaction.

Approach: measure the threat → understand why customers switch → options → choose by margin and strategic segment.

Model answer: "Don't react instantly. Check churn and win/loss by segment, interview switchers, and see whether price or features drive it. Options: hold price and sharpen value messaging; launch a lighter tier for price-sensitive users; bundle; raise switching value ethically. Choose based on margin impact and which segment matters strategically."

Assumptions & trade-offs: price cuts are hard to reverse; a light tier can cannibalise.

Common mistakes: automatic price cut; ignoring segments.

60-second answer: measure → understand → targeted response, not a blanket price cut.

ST5 · Hard — Should a music-streaming service expand into podcasts or audiobooks?

What it tests: platform expansion logic, economics, differentiation.

Approach: reframe the job → benefits → costs → strategy → metrics. *(Hypothetical analysis.)*

Model answer: "Frame it as expanding from 'music' to 'audio companionship'. Benefits: more listening time, differentiation, ad inventory. Costs: content and rights, discovery complexity. Strategy: invest in discovery and creator tools rather than only exclusives. Measure incremental listening time and retention, not catalogue size."

Assumptions & trade-offs: exclusives are expensive; cluttered UX for music-only users.

Common mistakes: measuring success by catalogue size.

60-second answer: job expansion → discovery-led strategy → incremental listening and retention.

ST6 · Medium — Build, buy or partner for identity verification?

What it tests: build/buy/partner reasoning.

Approach: is it core? → speed → total cost → compliance → lock-in.

Model answer: “Buy or partner: identity verification is regulated and commodity-like, and speed matters. Keep control of the onboarding UX and data handling, check vendor compliance and uptime, build an abstraction layer to switch vendors, and revisit building only if scale economics or differentiation justify it.”

Assumptions & trade-offs: vendor cost at scale; data residency.

Common mistakes: building commodity capabilities; ignoring exit costs.

60-second answer: commodity + regulated → partner, with an abstraction layer and a review point.

ST7 · Medium — What is the moat for a ride-hailing company?

What it tests: understanding of network effects and their limits.

Approach: list candidate moats → test each → conclude on durability.

Model answer: “Local network effects (density cuts wait times), brand trust and safety record, data for pricing and ETAs, and driver relationships. Weaknesses: riders and drivers multi-home, and regulation. So the moat is local and must be continuously defended with reliability and driver economics.”

Assumptions & trade-offs: subsidies by rivals can erode density quickly.

Common mistakes: saying “network effects” without testing them.

60-second answer: local density + trust + data; weakened by multi-homing; defend city by city.

ST8 · Hard — How would you set a 3-year vision and strategy for a B2B SaaS product?

What it tests: long-range thinking, sequencing, measurable strategy.

Approach: vision → beachhead → adjacent segments → product pillars → GTM motion → metrics → what we won't do.

Model answer: "Vision: the customer future (e.g., 'every team closes the loop on decisions'). Strategy: win a beachhead, then adjacent segments; three product pillars; integrations as a platform; GTM from PLG to sales-assist; pricing evolution; an NSM with guardrails; explicit 'won't do' list. Validate yearly with retention, NRR and win/loss."

Assumptions & trade-offs: focus vs breadth; platform investment vs features.

Common mistakes: a feature list dressed up as strategy.

60-second answer: vision → beachhead → pillars → GTM → NSM → annual validation.

Execution (8 questions)

EX1 · Medium — Your launch is slipping by three weeks. What do you do?

What it tests: ownership, communication, trade-offs.

Approach: root cause → options (scope, date, resources) → recommend → communicate → record.

Model answer: "Find out why — scope growth, a dependency or quality issues. Options: cut scope to keep the date, add people (rarely helps late), or move the date. Choose by cost of delay and quality risk. Tell stakeholders immediately with impact and the new plan; update the decision log."

Assumptions & trade-offs: some dates are contractual; quality cuts create debt.

Common mistakes: hiding the slip; promising dates for engineers.

60-second answer: root cause → scope-vs-date options → recommendation → early communication.

EX2 · Medium — How do you write good user stories and acceptance criteria?

What it tests: requirement clarity and testability.

Approach: INVEST stories → Given/When/Then criteria → non-functional and analytics needs → Ready/Done.

Model answer: "Stories describe user value: 'As a first-time buyer, I want to see my estimated total in the cart so I can decide before paying.' Criteria cover the happy path, edge cases, errors, accessibility and analytics events, in Given/When/Then form. I review them with engineering and QA before sprint planning." *(Module 6.)*

Assumptions & trade-offs: detail vs flexibility; documentation overhead.

Common mistakes: solution-prescriptive stories; untestable criteria ("fast", "intuitive").

60-second answer: INVEST + Given/When/Then + states + events, reviewed with the team.

EX3 · Medium — Engineering says a feature will take 3 months; leadership wants it in 1.

What it tests: negotiation and scope slicing.

Approach: understand drivers → slice → defer risk → options with trade-offs → joint escalation.

Model answer: "Understand what drives the estimate; find the smallest valuable slice; defer non-essential risk behind a flag; present options (1-month slice, 3-month full) with trade-offs; never commit dates on engineers' behalf. If needed, escalate jointly with an options paper."

Assumptions & trade-offs: a slice may need rework later.

Common mistakes: pressuring engineers; promising leadership the impossible.

60-second answer: slice + options + joint decision.

EX4 · Hard — A production incident is causing payment failures. What is your role as PM?

What it tests: incident leadership without overstepping.

Approach: support the incident lead → quantify impact → communicate → remediate customers → review.

Model answer: "Engineering leads the fix. I quantify user and business impact, coordinate communications (support, status page, leadership), decide customer remediation (refunds, messaging), and ensure a blameless post-incident review whose prevention items get prioritised." *(Extra topic 13: don't declare it over just because the dashboard looks better briefly.)*

Assumptions & trade-offs: speed vs accuracy of communications.

Common mistakes: debugging instead of coordinating; skipping the review.

60-second answer: impact, comms, remediation, review.

EX5 · Medium — How do you decide when a feature is ready to launch?

What it tests: quality bar and launch discipline.

Approach: acceptance → quality gates → measurement → support → rollout plan.

Model answer: "Acceptance criteria met; QA/UAT, performance, accessibility, security and privacy reviews done; tracking validated; support ready; rollout plan with flag and rollback owner; success and guardrail thresholds defined."

Assumptions & trade-offs: perfect vs good enough for a staged rollout.

Common mistakes: launching without measurement or rollback.

60-second answer: gates + measurement + rollback + support.

EX6 · Medium — How do you manage dependencies across teams?

What it tests: coordination and risk management.

Approach: map early → agree interfaces → track → sequence → escalate with options.

Model answer: "Map dependencies with owners, dates and risks; agree interface contracts; track them on a shared board; sequence work to unblock the critical path; hold short syncs; escalate early with options when blocked."

Assumptions & trade-offs: coordination overhead vs surprise risk.

Common mistakes: discovering dependencies at integration time.

60-second answer: map, contract, track, sequence, escalate early.

EX7 · Easy — What is the difference between Definition of Ready and Definition of Done?

What it tests: Agile fundamentals.

Approach: define both → example → why both matter.

Model answer: "Ready = criteria a team may use before starting an item (clear, estimated, testable). Done = shared quality criteria for completed work (tested, reviewed, documented, deployed per team rules)." *(Lecture 22.)*

Assumptions & trade-offs: an over-strict Ready can block flow.

Common mistakes: treating them as the same checklist.

60-second answer: Ready = can start; Done = truly finished.

EX8 · Hard — After launch, adoption is low. What do you do?

What it tests: diagnosis and willingness to iterate or stop.

Approach: measurement → exposure → segments → interviews → decide (improve, reposition, sunset).

Model answer: “Check tracking; check exposure (did people even see it?); segment by user type; interview users who saw it but didn't use it; compare with the original hypothesis. Then improve onboarding or placement, iterate on value, or sunset. Low use alone isn't enough to remove a feature — persistent low value is.”

Assumptions & trade-offs: sunk cost; maintenance cost of unused features.

Common mistakes: adding promotions without understanding why.

60-second answer: verify → exposure → why → decide.

Product analytics (10 questions)

AN1 · Easy — What metrics would you track for a food-delivery app?

What it tests: metric hierarchy and guardrails.

Approach: North Star → inputs → economics → guardrails.

Model answer: “North Star: completed, problem-free orders per active customer per month. Inputs: activation, order frequency, cart conversion, payment success, on-time delivery. Economics: AOV, contribution per order, CAC payback. Guardrails: refunds, complaints, cancellations.” *(Module 7.)*

Assumptions & trade-offs: “problem-free” needs a precise definition.

Common mistakes: listing 20 unrelated metrics; vanity metrics like total downloads.

60-second answer: NSM + 4 inputs + 2 economics + 3 guardrails.

AN2 · Medium — Define activation for a team-collaboration product.

What it tests: linking an early behaviour to long-term value.

Approach: candidate events → correlation with retention → causal test.

Model answer: "An event reflecting team value that predicts retention, e.g., 'a workspace with ≥ 3 members exchanging messages in a channel within 7 days.' Validate by comparing retention of activated vs not, then run an experiment to see whether helping teams reach it improves retention."

Assumptions & trade-offs: correlation ≠ causation; too-easy activation events are meaningless.

Common mistakes: using sign-up as activation.

60-second answer: team-level value event, validated against retention, tested causally.

AN3 · Medium — An A/B test shows +3% conversion with p = 0.04. Ship?

What it tests: experiment hygiene beyond the p-value.

Approach: pre-registration → power and duration → SRM → guardrails → CI and practical value → novelty → decision.

Model answer: "Check: was this the pre-registered primary metric? Adequate power and full weeks? Any sample ratio mismatch? Guardrails flat? Does the low end of the confidence interval still justify the cost? Is there a novelty risk or a segment that reverses? If all good, ship with a holdout; if not, extend or rerun."

Assumptions & trade-offs: speed of shipping vs certainty.

Common mistakes: shipping on $p < 0.05$ alone; peeking.

60-second answer: p is necessary, not sufficient — check design, guardrails, CI and cost.

AN4 · Medium — Explain p-value and confidence interval to a non-technical stakeholder.

What it tests: clarity and correctness under simplification.

Approach: plain-language definitions → why the CI matters for decisions.

Model answer: "The p-value tells us how surprising our result would be if the change actually did nothing — small means unlikely to be pure chance. The confidence interval gives the range of effect sizes consistent with our data, like 'between +1 and +4 points'. It's the range that matters for the business decision."

Assumptions & trade-offs: simplification loses nuance; avoid saying "95% chance the variant is better".

Common mistakes: misdefining p as the probability the variant works.

60-second answer: the two sentences above.

AN5 · Medium — How would you calculate retention and what pitfalls exist?

What it tests: cohort thinking and definitional rigour.

Approach: cohort → active definition → window → formula → pitfalls.

Model answer: "Group users by signup week; define 'active' as a meaningful action; retention in week k = active users in week k ÷ cohort size. Pitfalls: mixed definitions, small cohorts, survivorship bias, seasonality, counting any app open as active."

Assumptions & trade-offs: rolling vs bracketed windows.

Common mistakes: averaging across different cohorts without care.

60-second answer: cohort ÷ active-in-window, with clear definitions and pitfalls.

AN6 · Hard — DAU grew 10% but revenue fell 5%. Explain possible reasons.

What it tests: decomposition and segment thinking.

Approach: revenue = active users × payer conversion × ARPPU → check each by segment.

Model answer: "Mix shift (new users are free or low spenders), pricing/promotion changes, payment failures, lower conversion of engaged users, refunds, or measurement changes. Decompose revenue = active users × payer conversion × ARPPU and check each component by segment and date."

Assumptions & trade-offs: some reasons are healthy (deliberate acquisition of free users).

Common mistakes: jumping to one story without decomposing.

60-second answer: decompose, segment, check payments and pricing.

AN7 · Medium — Write SQL for conversion from checkout start to payment.

What it tests: practical SQL and definitional awareness.

Approach: count distinct users per step → ratio → note funnel strictness.

Model answer:

```
SELECT ROUND(100.0 * COUNT(DISTINCT CASE WHEN event_name='payment_success' THEN
       / COUNT(DISTINCT CASE WHEN event_name='checkout_start' THEN user_id END)
FROM events;
```

On the synthetic dataset this returns 284 ÷ 322 = **88.2%** (users who ever reached each step).

Assumptions & trade-offs: a strict funnel needs ordering and a time window per user.

Common mistakes: counting events instead of users; dividing by the wrong step.

60-second answer: distinct users at each step, ratio, mention ordering/time windows.

AN8 · Medium — Correlation vs causation — give a product example.

What it tests: causal reasoning.

Approach: example → confounder → how to test.

Model answer: “Users of feature X retain better, but power users may both use X and retain for other reasons (confounding). Test with a randomised experiment or a credible quasi-experiment. Simpson's paradox shows how an uneven traffic mix can reverse conclusions.”

Assumptions & trade-offs: experiments aren't always possible; use holdouts or staggered rollouts.

Common mistakes: claiming a feature “drives” retention from a correlation.

60-second answer: example → confounder → experiment.

AN9 · Hard — How would you measure the success of a recommendation system?

What it tests: online vs offline evaluation and long-term thinking.

Approach: offline metrics → online metrics → long-term holdout → guardrails.

Model answer: “Offline: precision/recall@k on historical data. Online: click-through and, more importantly, downstream outcomes (purchases, listening time, satisfaction) via A/B tests; diversity and novelty; long-term retention via holdout. Guardrails: complaints, returns, filter bubbles, creator/seller fairness.”

Assumptions & trade-offs: clicks vs satisfaction; short vs long term.

Common mistakes: optimising CTR alone.

60-second answer: offline + online outcomes + holdout + guardrails.

AN10 · Medium — What would you put on a PM dashboard for a new feature?

What it tests: dashboard design tied to decisions.

Approach: exposure → adoption → outcome → guardrails → segments → context.

Model answer: "Exposure (who saw it), adoption (who used it), the primary outcome, guardrails, segment cuts, release annotations, data freshness and metric definitions."

Assumptions & trade-offs: too many charts dilute focus.

Common mistakes: no definitions; no guardrails.

60-second answer: the list above, one headline per question.

Estimation / guesstimates (8 questions)

ES1 · Medium — Estimate daily ride-hailing trips in a large Indian metro.

What it tests: structure, assumptions, sanity checks.

Approach: clarify definitions → top-down → bottom-up → reconcile → range.

Model answer: "Top-down: 20M population → 14M adults → 8% weekly users = 1.12M × 3 trips/week ≈ 480k/day. Bottom-up: 55,000 active drivers × 9 trips ≈ 495k/day. Range 450k–500k; sanity check ≈ 3.4% of adults ride daily." *(Module 3; course Lecture 7 — practice figures, not company data.)*

Assumptions & trade-offs: share of weekly users and trips per driver are the most uncertain.

Common mistakes: inconsistent definitions (cancelled rides); no range.

60-second answer: two methods, reconciled, range, key assumptions.

ES2 · Medium — How many food-delivery orders happen per day in a city of 10 million?

What it tests: segmentation and supply-side checks.

Approach: households → eligible → frequency → adjust → supply check.

Model answer: "10M ÷ 4 = 2.5M households; 60% with smartphones in delivery zones = 1.5M; 30% order at least weekly = 450k; 2.5 orders/week = 1.125M/week ≈ 160k/day; +20% for offices/single residents ≈ 190k/day. Range 150k–250k. Supply check: 15,000 couriers × 12 orders/day = 180k ✓."

Assumptions & trade-offs: household size and frequency vary by city.

Common mistakes: skipping the supply check.

60-second answer: ≈ 190k/day (150k–250k) with demand and supply methods.

ES3 · Easy — How many school lunches are bought daily at a 1,000-student school?

What it tests: basic decomposition.

Approach: students × attendance × buying rate.

Model answer: "1,000 × 60% at school × 50% buying = 300 lunches (course example). Discuss day-of-week and seasonal variation."

Assumptions & trade-offs: attendance and buying rates.

Common mistakes: no stated assumptions.

60-second answer: 300, with the three assumptions named.

ES4 · Medium — Estimate the annual revenue of a coffee chain with 200 outlets.

What it tests: revenue modelling.

Approach: per-outlet daily revenue → annualise → scale → range.

Model answer: "400 cups/day × ₹220 average ticket = ₹88,000; + 25% food ≈ ₹1.1 lakh/day; × 360 days ≈ ₹3.96 crore per outlet; × 200 ≈ ₹790 crore/year. Range ₹600–950 crore."

Assumptions & trade-offs: outlet mix (airport vs suburb) changes footfall a lot.

Common mistakes: treating all outlets as identical without saying so.

60-second answer: ≈ ₹790 crore with per-outlet logic and a range.

ES5 · Medium — How many smartphones are sold in India per year?

What it tests: replacement-cycle reasoning.

Approach: installed base ÷ replacement cycle + new users − second-hand.

Model answer: "Assume ~700M users; ~3-year replacement → ~230M; + ~15M new users ≈ 245M; −30% for second-hand and multi-device effects ≈ 170M new devices (range 140–190M). Check against industry reports in real life."

Assumptions & trade-offs: user base and replacement cycle are the biggest levers.

Common mistakes: population × penetration without replacement logic.

60-second answer: base ÷ cycle + new − second-hand ≈ 170M.

ES6 · Hard — Estimate yearly image storage for a messaging app.

What it tests: technical estimation.

Approach: DAU × senders × images × size → annualise → replication.

Model answer: "100M DAU × 20% sending × 5 images × 0.2 MB = 20 TB/day ≈ 7.3 PB/year before replication; ×3 replication ≈ 22 PB. Retention and deletion policies matter."

Assumptions & trade-offs: compression, media expiry, end-to-end encryption effects.

Common mistakes: forgetting replication and retention.

60-second answer: ≈ 7 PB/year raw, ~22 PB replicated.

ES7 · Medium — How many EV chargers does a city need?

What it tests: demand-to-capacity reasoning.

Approach: EVs × share without home charging × sessions/week ÷ sessions per charger/week → peak adjustment.

Model answer: "Example: 100,000 EVs × 40% without home charging × 2 sessions/week = 80,000 sessions/week; one fast charger can handle ~10 sessions/day ≈ 70/week → ~1,150 chargers; add 30% for peaks and geography ≈ 1,500 (range 1,200–2,000)."

Assumptions & trade-offs: charger speed mix; utilisation targets.

Common mistakes: ignoring peaks and location coverage.

60-second answer: sessions needed ÷ sessions per charger, plus peak buffer.

ES8 · Easy — Estimate the number of piano tuners in a city.

What it tests: the classic Fermi method.

Approach: pianos × tunings/year ÷ tunings per tuner per year.

Model answer: "A city of 5M with 2M households; 2% with pianos = 40,000 + 2,000 institutional; tuned once a year = 42,000 tunings; a tuner does ~4/day × 250 days = 1,000/year → about 40–45 tuners."

Assumptions & trade-offs: piano ownership varies widely by city.

Common mistakes: chasing the "right" number instead of showing method.

60-second answer: ≈ 40 tuners with explicit assumptions.

Root cause analysis (8 questions)

RC1 · Medium — Daily orders dropped 15% yesterday. What do you do?

What it tests: structured diagnosis.

Approach: clarify the metric and window → data integrity → internal changes → external factors → segment → hypotheses → verify.

Model answer: "First confirm it's real — did order *events* fall but the orders table didn't? Compare with the same day last week. Segment by platform, city, new/returning, payment method and restaurant. If only Android in one city dropped, check the latest release and payment-gateway logs there. Rank hypotheses by likelihood × impact and check the cheapest first." *(Lecture 22: ask exactly which metric fell before an RCA.)*

Assumptions & trade-offs: speed of diagnosis vs thoroughness.

Common mistakes: jumping to a cause; skipping data-quality checks.

60-second answer: real? → when/where/who → internal vs external → hypotheses → verify.

RC2 · Medium — The app-store rating fell from 4.4 to 4.1. Why?

What it tests: qualitative + quantitative investigation.

Approach: review mining by date/version → crash data → policy changes → rating-prompt changes.

Model answer: "Tag reviews by date and app version; correlate with releases; check crash rates; check whether a policy change (e.g., fees) triggered anger; check whether rating prompts changed (sampling bias). Fix the top cause and monitor."

Assumptions & trade-offs: ratings lag reality; vocal minorities.

Common mistakes: responding to reviews without diagnosing.

60-second answer: reviews by version + crashes + policy + prompt changes.

RC3 · Hard — Sign-ups up 20%, activation down 8 points. Explain.

What it tests: mix-shift reasoning.

Approach: segment activation by channel and cohort → check onboarding and tracking changes.

Model answer: "Likely mix shift — a new channel bringing lower-intent users; or an onboarding change; or a change to the activation event's tracking; or a bug for one segment. If a new channel explains it, the funnel isn't broken — the traffic is different, and channel ROI should be judged on activated users."

Assumptions & trade-offs: volume vs quality of acquisition.

Common mistakes: "fix onboarding" without segmenting.

60-second answer: segment by channel/cohort; mix shift is the first suspect.

RC4 · Medium — Session length increased but retention decreased. Is that good?

What it tests: metric interpretation.

Approach: possible meanings → checks.

Model answer: "Not necessarily. Users may be struggling (tasks taking longer), or a minority binges while others leave. Check task success, segment by cohort, look at satisfaction and support tickets."

Assumptions & trade-offs: engagement vs efficiency products differ.

Common mistakes: celebrating longer sessions.

60-second answer: longer may mean harder; check success and cohorts.

RC5 · Medium — Ride cancellations spiked in one city.

What it tests: marketplace diagnosis.

Approach: supply → events/weather → pricing → app versions → ETA accuracy → payment modes.

Model answer: "Check drivers online vs demand, local events or weather, pricing changes, driver-app bugs, ETA accuracy and payment-mode mismatches; segment by time and area; talk to operations."

Assumptions & trade-offs: rider vs driver cancellations need separate analysis.

Common mistakes: treating all cancellations as one metric.

60-second answer: split by who cancelled, when, where; check supply and changes.

RC6 · Medium — Payment success rate fell from 95% to 91%.

What it tests: technical-business diagnosis.

Approach: split by method, issuer, error code, app version → provider status → mitigation.

Model answer: "Split by card/UPI/wallet, issuer/bank, error code and app version; check the gateway status and recent changes (3-D Secure flow, SDK update); contact the provider; mitigate with routing or retries; communicate with support."

Assumptions & trade-offs: some failures are customer-side (insufficient funds).

Common mistakes: assuming it's always the gateway.

60-second answer: decompose by method/issuer/error → recent changes → mitigate.

RC7 · Hard — In a community product, group creation is down. Diagnose.

What it tests: defining meaningful metrics and community dynamics.

Approach: define the metric → measurement → segment → discovery/moderation/competition hypotheses → test.

Model answer: "Define the metric (new groups that become active?), check measurement, segment by country, platform and entry point; examine discovery changes, spam/moderation enforcement changes, and competition. Test hypotheses with data and interviews." *(Course Lecture 36 case — hypotheses, not facts about any company.)*

Assumptions & trade-offs: fewer but better groups may be fine.

Common mistakes: optimising raw group counts.

60-second answer: define meaningful creation → segment → hypotheses → test.

RC8 · Medium — Customer complaints doubled after a redesign.

What it tests: owning a regression.

Approach: read and tag → identify flows → usability test → accessibility check → fix or roll back.

Model answer: "Tag complaints to flows; run a quick usability test on the new design; check for accessibility regressions; consider rollback or targeted fixes; compare task success before and after."

Assumptions & trade-offs: change aversion vs real problems — look at behaviour, not only sentiment.

Common mistakes: dismissing complaints as change aversion.

60-second answer: tag → test → fix or roll back → measure.

Prioritization (7 questions)

PR1 · Easy — Which prioritisation frameworks do you use?

What it tests: knowing when each framework fits.

Approach: decision type → framework → limitation.

Model answer: "It depends on the decision: RICE for comparable features, MoSCoW for release scope, WSJF when timing matters, Kano for customer expectations, weighted scoring for strategic bets. I always add a sensitivity check and dependencies, and take mandatory work first."

Assumptions & trade-offs: speed vs rigour.

Common mistakes: listing frameworks without when/why.

60-second answer: framework-per-decision + sensitivity + mandatory work first.

PR2 · Medium — You have 10 feature requests and capacity for 3. Walk me through it.

What it tests: applied prioritisation and communication.

Approach: goal → group by problem → size → estimate → score → dependencies/risks → choose → communicate.

Model answer: "Clarify the goal; group requests by underlying problem (often 10 requests are 4 problems); size reach and impact with data; get effort estimates from engineering; score with RICE; check dependencies, risks and strategy; pick three; tell requesters what's not happening and why, with a review date."

Assumptions & trade-offs: estimates are uncertain; politics.

Common mistakes: scoring requests instead of problems; silent rejections.

60-second answer: problems not requests → score → sequence → explain.

PR3 · Medium — Compute RICE: reach 5,000, impact 2, confidence 50%, effort 2.

What it tests: arithmetic and interpretation.

Approach: compute → interpret → how to improve confidence.

Model answer: "5,000 × 2 × 0.5 ÷ 2 = 2,500. I'd confirm reach is per quarter and consider a cheap test to raise confidence before committing two person-months."

Assumptions & trade-offs: scales are team conventions.

Common mistakes: putting effort in the numerator; mixing time periods.

60-second answer: 2,500, then raise confidence cheaply.

PR4 · Hard — Tech debt vs new features: how do you decide?

What it tests: long-term thinking and quantifying invisible work.

Approach: quantify debt cost → baseline capacity → treat big items as investments → SLO/error-budget rule.

Model answer: "Quantify tech debt's cost — incident frequency, slower delivery, error rates, features blocked. Reserve baseline capacity for reliability, and treat large debt items as investments with expected outcomes (e.g., halve checkout incidents). If SLOs are breached and the error budget is spent, reliability work takes priority."

Assumptions & trade-offs: short-term features vs long-term velocity.

Common mistakes: "we'll fix it later" forever; or rewriting everything.

60-second answer: quantify, reserve capacity, use error budgets.

PR5 · Medium — A big customer threatens to churn unless you build X.

What it tests: B2B judgement and avoiding one-off customisations.

Approach: understand need and ARR at risk → breadth of demand → options → cost of delay → decide with leadership.

Model answer: "Understand the need and the ARR at risk; check how many other customers need it; explore workarounds, smaller scope or a timeline; compare with the cost of delaying displaced work; decide with leadership, avoiding one-off features that fragment the product."

Assumptions & trade-offs: revenue now vs product coherence.

Common mistakes: saying yes reflexively; or dismissing revenue risk.

60-second answer: need, breadth, options, explicit trade-off decision.

PR6 · Medium — How do you prioritise bugs?

What it tests: severity reasoning.

Approach: severity × frequency × reach → special classes (security, data loss) → SLAs.

Model answer: “Rank by severity (impact on users, data, money, safety) × frequency × reach; security and data-loss bugs first; agree SLAs per severity; batch low-severity fixes; track escaped defects to improve testing.”

Assumptions & trade-offs: fix vs feature capacity.

Common mistakes: first-in-first-out bug queues.

60-second answer: severity-based with SLAs.

PR7 · Hard — How do you decide what goes into an MVP?

What it tests: learning-driven scoping.

Approach: riskiest assumption → learning goal and threshold → minimal safe scope → simulated vs real → next decision.

Model answer: “Identify the riskiest assumption; define what we need to learn and the success threshold; include only what's needed to test it safely; label what's simulated; plan the decision the result will drive. An MVP is not the same as a prototype or a public launch.”

Assumptions & trade-offs: too minimal can't test value; too big wastes time.

Common mistakes: “MVP = v1 with fewer features”.

60-second answer: riskiest assumption → smallest safe test → decision.

Technical PM (9 questions)

TE1 · Easy — What is an API? Explain to a non-technical person.

What it tests: clear technical communication.

Approach: analogy → real example → failure case.

Model answer: “An API is like a restaurant menu and waiter: it lists what you can ask for, you place a request in a standard format, and the kitchen sends back the dish — or an error if it's unavailable. Apps use APIs to ask other systems for data or actions, like a weather app requesting today's forecast.”

Assumptions & trade-offs: analogies simplify; mention contracts and errors.

Common mistakes: jargon-heavy definitions.

60-second answer: the analogy plus one example.

TE2 · Medium — What happens when you type a URL and press Enter?

What it tests: systems fundamentals.

Approach: DNS → TCP/TLS → HTTP → edge → server → response → render.

Model answer: “DNS resolves the name to an IP; the browser opens a TCP connection and TLS handshake; sends an HTTP request that may be served by a CDN or reach a load balancer and app servers, which query databases or services; the response returns with a status code; the browser renders and fetches assets.” *(Module 9.)*

Assumptions & trade-offs: caching changes the path.

Common mistakes: skipping DNS/TLS; no mention of failure points.

60-second answer: the chain above with latency/failure notes.

TE3 · Medium — REST vs GraphQL — when would you choose each?

What it tests: trade-off reasoning.

Approach: describe both → strengths/weaknesses → choose by use case.

Model answer: "REST for simple, cacheable resource APIs and public integrations; GraphQL for complex UIs that need flexible data from many sources in one request. Consider caching, rate limiting and query-cost control."

Assumptions & trade-offs: team expertise, tooling.

Common mistakes: "GraphQL is always better".

60-second answer: REST for simplicity/caching; GraphQL for flexible UI data.

TE4 · Medium — SQL vs NoSQL for an orders system?

What it tests: data-store selection.

Approach: access patterns → consistency needs → choose.

Model answer: "Relational (SQL) for orders and payments — transactions and consistency matter. NoSQL can suit catalogues or event logs at huge scale. Choose by access pattern and consistency needs, not fashion."

Assumptions & trade-offs: scale vs consistency.

Common mistakes: choosing NoSQL "for scale" without need.

60-second answer: SQL for money; NoSQL for flexible/large-scale non-transactional data.

TE5 · Hard — Design a notification system.

What it tests: requirements-first system design.

Approach: use cases and scale → components → data → failures → abuse/consent → metrics → rollout.

Model answer: "Producers publish events to a queue; a notification service applies preferences, frequency caps, templates and quiet hours; channel providers (push/SMS/email) deliver; receipts flow back. Add dedup, retries with back-off and fallbacks for critical messages; respect consent; measure delivery, opt-outs and complaints." *(Module 9.)*

Assumptions & trade-offs: transactional vs marketing priority; cost per channel.

Common mistakes: jumping to technologies; ignoring fatigue and consent.

60-second answer: queue → central service with preferences/caps → providers → receipts → metrics.

TE6 · Medium — How do you prevent double charges when a payment request is retried?

What it tests: idempotency and reconciliation.

Approach: idempotency keys → webhook dedup → reconciliation → honest UX.

Model answer: "Send an idempotency key with each charge so retries return the original result; dedupe webhooks by event ID; reconcile with the provider daily; show 'payment pending' honestly when status is unknown."

Assumptions & trade-offs: provider support for idempotency.

Common mistakes: retrying blindly.

60-second answer: idempotency key + dedup + reconciliation.

TE7 · Medium — Authentication vs authorisation?

What it tests: security fundamentals.

Approach: define both → example → status codes.

Model answer: "Authentication is who you are (login, OTP, passkeys); authorisation is what you're allowed to do (roles, scopes). A logged-in user is authenticated but still not authorised to see another user's order — the API should return 403 (or 404 to avoid revealing it exists)."

Assumptions & trade-offs: UX of re-authentication vs security.

Common mistakes: conflating the two.

60-second answer: who vs what, with the order example.

TE8 · Hard — API p95 latency doubled. How do you approach it as a PM?

What it tests: incident thinking and user-impact quantification.

Approach: confirm scope → correlate with changes → dependencies → user impact → mitigation → communication → review.

Model answer: "Confirm which endpoints and regions; correlate with releases and traffic; check dependencies (DB, third parties); quantify user impact (conversion by latency bucket); work with engineering on mitigation (rollback, cache, scale); communicate; run a post-incident review."

Assumptions & trade-offs: fast rollback vs root-cause understanding.

Common mistakes: watching averages only.

60-second answer: scope → cause candidates → impact → mitigate → review.

TE9 · Medium — Explain your product's architecture in an interview.

What it tests: honest technical fluency.

Approach: users → clients → gateway → services → data → third parties → core flow → failures → your decisions.

Model answer: "Walk the core journey through each component, say where failures happen and how we handle them, and separate what I decided (fallbacks, events, SLAs) from what engineering owned." *(Lectures 30 and 52: distinguish knowledge from assumptions; protect sensitive details.)*

Assumptions & trade-offs: depth depends on role.

Common mistakes: claiming technical decisions you didn't make.

60-second answer: a 6-box diagram narrated along the core flow.

AI product management (10 questions)

AI1 · Easy — When should you NOT use AI for a feature?

What it tests: judgement and non-AI baselines.

Approach: list conditions → baseline comparison.

Model answer: "When a simple rule or search works reliably; when errors are costly and can't be caught; when we lack authorised data; when latency or cost exceeds the value; when we can't evaluate quality. Always compare with a non-AI baseline."

Assumptions & trade-offs: AI may still help later as data matures.

Common mistakes: AI for its own sake.

60-second answer: the five conditions + baseline.

AI2 · Medium — Explain RAG vs fine-tuning.

What it tests: conceptual accuracy.

Approach: definitions → use cases → combination → limits.

Model answer: "RAG retrieves current, permitted knowledge at answer time — good for facts, citations and fast updates. Fine-tuning adjusts behaviour or format using training examples — good for consistent style or specialised tasks. Neither guarantees truth; they can be combined." *(Module 10.)*

Assumptions & trade-offs: cost and maintenance of each.

Common mistakes: calling document upload "fine-tuning".

60-second answer: knowledge vs behaviour.

AI3 · Medium — How would you evaluate an LLM-powered support bot before launch?

What it tests: evaluation rigour.

Approach: eval set → rubric → reviewers/judges → severity → gates → pilot.

Model answer: "Build a curated eval set (common, edge, adversarial, escalation cases); score correctness, groundedness, safety and policy compliance with a rubric; use human review and a validated LLM judge; weight failures by severity; set launch gates; pilot with human-in-the-loop; then track online metrics."

Assumptions & trade-offs: cost of human review; judge bias.

Common mistakes: demos instead of evals.

60-second answer: eval set + rubric + severity gates + pilot.

AI4 · Hard — Your AI feature hallucinates 3% of the time. Launch or not?

What it tests: risk-based decisions.

Approach: harm assessment → mitigations → routing → decision.

Model answer: "Depends on harm: 3% wrong recipe ideas may be fine with labels; 3% wrong policy answers in banking is not. Reduce with grounding and refusals, route high-risk intents to humans, show citations, and measure severity, not just rate."

Assumptions & trade-offs: coverage vs accuracy.

Common mistakes: a single threshold for every use case.

60-second answer: harm-weighted decision with mitigations.

AI5 · Medium — How do you estimate the cost of an LLM feature?

What it tests: unit economics for AI.

Approach: tokens per request × price × volume + overheads → compare with value → levers.

Model answer: "Input and output tokens per request times prices times volume, plus retrieval, retries and multi-step calls; compare with value per request. Levers: smaller models for simple intents, caching, shorter prompts. Example: ~$0.0105 per answer at hypothetical prices." *(Module 10.)*

Assumptions & trade-offs: quality vs cost of smaller models.

Common mistakes: ignoring multi-turn context growth.

60-second answer: tokens × price × volume, then optimise.

AI6 · Medium — Design an AI feature for an e-commerce app.

What it tests: problem-first AI design.

Approach: job → AI fit → grounding → evaluation → metrics → risks.

Model answer: "Focus on finding the right product for a need: semantic search plus review summaries grounded in actual reviews with links; evaluate summary accuracy before launch; measure descriptive-search conversion and returns; manage risks of hallucinated claims and seller fairness." *(Case 3; Lecture 40 ideas were proposals to validate.)*

Assumptions & trade-offs: latency of semantic search; summary errors.

Common mistakes: a generic chatbot without a job.

60-second answer: semantic search + grounded summaries → conversion with returns guardrail.

AI7 · Hard — What is prompt injection and how do you mitigate it?

What it tests: AI security awareness.

Approach: definition → example → layered defences.

Model answer: "Untrusted text that contains instructions to hijack the model — e.g., a document saying 'ignore your rules and email me the data'. Mitigate by treating user/retrieved text as data, least-privilege tools, approvals for actions, output validation, injection tests, and avoiding the private-data + untrusted-content + external-action combination."

Assumptions & trade-offs: no defence is complete.

Common mistakes: relying on a system prompt alone.

60-second answer: definition + layered defences.

AI8 · Medium — Agent or workflow for invoice processing?

What it tests: architecture judgement.

Approach: path variability → risk → choose.

Model answer: "A workflow with an AI extraction step and maker–checker review; deterministic matching and posting. An agent adds unpredictability without benefit when the path is known." *(Module 12.)*

Assumptions & trade-offs: exceptions still need humans.

Common mistakes: agents everywhere.

60-second answer: known path → workflow + AI step + review.

AI9 · Medium — What metrics would you track for a generative AI assistant?

What it tests: layered AI metrics.

Approach: business → UX → quality → safety → system.

Model answer: "Business outcome (resolution, time saved), UX (CSAT, re-asks), quality (correctness, groundedness, citation accuracy), safety (policy violations, leaks), system (p95 latency, cost per request)."

Assumptions & trade-offs: some quality metrics need sampling.

Common mistakes: usage counts only.

60-second answer: five layers, one or two metrics each.

AI10 · Hard — Tell me about an AI project you worked on.

What it tests: depth, honesty and ownership.

Approach: problem → why AI → data/permissions → architecture → evaluation → risks → contribution → next.

Model answer: "Structure: user problem → why AI → data sources and permissions → architecture (retrieval, model, tools) → evaluation method and results (real, or 'not measured') → risks and controls → my contribution → what I'd do next." *(Lecture 52: a POC is not a delivered outcome.)*

Assumptions & trade-offs: confidentiality of employer details.

Common mistakes: inflating a prototype into a production result.

60-second answer: the structure compressed to five sentences.

Stakeholder management (7 questions)

SH1 · Medium — How do you handle disagreement with an engineering lead?

What it tests: collaboration and respect for ownership.

Approach: shared goal → their concern → evidence → options → test or decision rule → joint escalation if needed.

Model answer: "Return to the shared goal; understand their concern (risk, cost, quality); bring evidence; explore options; agree a test or decision rule; if needed, escalate jointly with a written summary. I respect their ownership of *how* while I own the problem and priority." *(Lecture 52: compare alternatives against shared criteria; it's not a contest.)*

Assumptions & trade-offs: time spent aligning vs speed.

Common mistakes: overruling; or deferring on product decisions.

60-second answer: goal → concern → evidence → test → document.

SH2 · Medium — Sales keeps promising features to customers. What do you do?

What it tests: process design and relationships.

Approach: understand deals → give useful roadmap visibility → agree rules → create a request channel → close the loop.

Model answer: "Understand the deals at stake; share roadmap themes rather than dates; agree that commitments need PM sign-off; give sales a fast way to log requests with customer evidence; close the loop on decisions so they trust the process."

Assumptions & trade-offs: some strategic deals justify exceptions — decided explicitly.

Common mistakes: blaming sales; no feedback loop.

60-second answer: themes not dates + sign-off rule + request channel.

SH3 · Medium — How do you say 'no' to a senior leader?

What it tests: courage with tact.

Approach: acknowledge goal → trade-offs with data → alternatives → their decision if it's theirs → document.

Model answer: "Acknowledge the goal; show what we'd stop to do this, with data; offer alternatives (smaller version, later slot, quick test); if the decision is theirs, ask them to make it with the trade-off visible; document it."

Assumptions & trade-offs: relationship vs focus.

Common mistakes: a flat 'no'; or a silent 'yes'.

60-second answer: "Yes, if we stop X — here are the options."

SH4 · Hard — Two VPs want conflicting things from your team.

What it tests: navigating politics with structure.

Approach: clarify goals → map to company priorities → quantify options → recommendation → single approver → document.

Model answer: "Clarify each VP's goal; map both to company priorities; quantify options; propose a recommendation; bring both to one decision meeting with a single approver (DACI); record the decision."

Assumptions & trade-offs: the approver may be above both.

Common mistakes: trying to satisfy both with half-measures.

60-second answer: shared yardstick + one approver.

SH5 · Medium — How do you keep stakeholders informed without overwhelming them?

What it tests: communication design.

Approach: map stakeholders → tier communications → no surprises.

Model answer: "Stakeholder map; tiered communications — a weekly one-pager for leadership, channel updates for partners, release notes for the wider organisation; and early warnings on risks."

Assumptions & trade-offs: frequency vs noise.

Common mistakes: one giant status email for everyone.

60-second answer: right message, right audience, right cadence.

SH6 · Medium — How do you work with designers?

What it tests: partnership mindset.

Approach: problems and evidence → early involvement → shared metrics → specific feedback → tests.

Model answer: "Bring problems and evidence, not solutions; involve designers in discovery; agree success metrics; give specific, user-centred feedback; protect time for edge states and accessibility; decide together what to test." *(Module 5.)*

Assumptions & trade-offs: speed vs exploration time.

Common mistakes: handing over wireframes as requirements.

60-second answer: problem + evidence + shared metrics + tests.

SH7 · Medium — How do you handle a stakeholder who keeps changing requirements?

What it tests: scope management.

Approach: understand why → change process → timeboxed discovery → sprint protection → transparent re-planning.

Model answer: "Understand why (new information or an unclear goal); agree a change process with a decision log and impact assessment; timebox discovery to settle unknowns; protect the current sprint's scope; renegotiate dates transparently."

Assumptions & trade-offs: some change is healthy learning.

Common mistakes: absorbing every change silently.

60-second answer: root cause + change process + visible impact.

Behavioural (8 questions)

Integrity rule

Use real stories. If you're transitioning into PM, use relevant experiences from your current role or course projects and label them accurately. Never invent metrics or titles (your course notes repeatedly flag this).

BE1 · Easy — Tell me about yourself.

What it tests: clarity, relevance, motivation.

Approach: present → past → future, ~90 seconds.

Model answer (template): "I'm currently a [role] at [company], where I [product-relevant responsibility]. Before that, I [relevant experience]. I've been deepening PM skills through a 55-session product course and portfolio projects — for example, [project] where I [specific contribution]. I'm excited about this role because [specific reason tied to the company's product]."

Assumptions & trade-offs: tailor to the role; depth vs brevity.

Common mistakes: reciting your whole CV; generic motivation.

60-second answer: the template, trimmed.

BE2 · Medium — Tell me about a time you influenced without authority.

What it tests: influence, data use, collaboration.

Approach: STAR with your specific actions.

Model answer (outline): "S: the team wanted to build a large client's request; I believed checkout reliability mattered more. T: align sales, engineering and my manager. A: I quantified lost orders from funnel and support data, understood the client's underlying need, proposed a two-week workaround, and presented options with a recommendation. R: we fixed payment retries first; state the real measured change — or say it wasn't measured — and the client outcome." *(Module 6.)*

Assumptions & trade-offs: be honest about what others did.

Common mistakes: describing team actions as your own; invented numbers.

60-second answer: S-T-A-R in four sentences.

BE3 · Medium — Tell me about a failure.

What it tests: self-awareness and learning.

Approach: real failure → your part → learning → what changed afterwards.

Model answer (structure): "Describe a real failure you contributed to (e.g., shipping without a tracking plan, so you couldn't evaluate impact); own your part; what you learned; and evidence you changed behaviour (you now require a tracking plan in every PRD)."

Assumptions & trade-offs: choose a meaningful but not catastrophic failure.

Common mistakes: fake failures ("I work too hard"); blaming others.

60-second answer: failure → my part → lesson → changed practice.

BE4 · Medium — Tell me about a time you used data to change a decision.

What it tests: analytical influence.

Approach: decision before → data and its limits → presentation → outcome → reflection.

Model answer (structure): "The team planned X; I analysed Y (be specific about source and limits); I presented it as options with trade-offs; the decision changed to Z; the outcome was [real result or 'not measured']; next time I'd [improvement]."

Assumptions & trade-offs: acknowledge uncertainty in the data.

Common mistakes: presenting data as certainty.

60-second answer: before → data → change → result.

BE5 · Medium — Describe a conflict with a teammate and how you resolved it.

What it tests: maturity and empathy.

Approach: situation → interests not positions → actions → outcome → relationship after.

Model answer (structure): "Explain the disagreement neutrally; how you understood their interests; the options you explored; how you decided (test, data or escalation); the outcome and how the relationship improved."

Assumptions & trade-offs: don't villainise the other person.

Common mistakes: a story where you were simply right.

60-second answer: interests → options → decision → relationship.

BE6 · Medium — A decision you made with incomplete information.

What it tests: decision-making under uncertainty.

Approach: known/unknown → cheap uncertainty reduction → reversible vs irreversible → decide → follow-up.

Model answer (structure): "What you knew and didn't; how you reduced uncertainty cheaply (a few interviews, a quick data pull); whether the decision was reversible; what you decided and the checkpoint you set; what happened."

Assumptions & trade-offs: speed vs certainty.

Common mistakes: waiting for perfect data; or no follow-up.

60-second answer: reduce uncertainty cheaply, decide, set a checkpoint.

BE7 · Medium — Why product management?

What it tests: genuine motivation and realistic expectations.

Approach: pattern from your experience → what PM involves → why this company.

Model answer: "Connect a real pattern from your work — noticing customer problems, coordinating teams, using data — to what PMs do; show you understand the hard parts (trade-offs, saying no, ambiguity); and tie it to this product."

Assumptions & trade-offs: honesty about gaps you're closing.

Common mistakes: "I want to be the CEO of the product."

60-second answer: pattern → understanding → fit.

BE8 · Hard — Tell me about the product you're most proud of.

What it tests: ownership, depth and outcomes.

Approach: problem and users → your role → key decisions and trade-offs → result → learning.

Model answer (structure): "Problem and users; your specific role; two key decisions and their trade-offs; results (real numbers or honest qualitative evidence); what you learned. For career switchers, a portfolio project clearly labelled as such."

Assumptions & trade-offs: confidentiality — share structure, not secrets.

Common mistakes: feature list without decisions.

60-second answer: problem → decisions → result → learning.

Leadership (6 questions)

LE1 · Medium — How do you set a vision for your team?

What it tests: strategic leadership and inclusion.

Approach: customer future → link to company strategy → co-create → translate → revisit.

Model answer: "Describe a customer-centred future state, tied to company strategy, co-created with the team; translate it into strategy, outcomes and a roadmap; revisit it when evidence changes."

Assumptions & trade-offs: inspiration vs concreteness.

Common mistakes: a slogan with no strategy beneath it.

60-second answer: customer future → strategy → outcomes.

LE2 · Hard — How do you build an outcome-focused culture in a feature-factory organisation?

What it tests: change leadership.

Approach: start small → prove → change rituals → OKRs with guardrails → share learning → sponsor.

Model answer: "Start with one team and one outcome, show results, change rituals (review outcomes, not outputs), introduce OKRs with guardrails, share learnings including failures, and enlist an executive sponsor."

Assumptions & trade-offs: delivery commitments still exist; transition gradually.

Common mistakes: mandating a framework top-down without proof.

60-second answer: pilot → proof → rituals → scale.

LE3 · Medium — How do you develop junior PMs?

What it tests: coaching.

Approach: expectations → ownership → feedback → pairing → scope growth.

Model answer: "Set clear expectations; give ownership of a problem area; regular SBI feedback; pair on discovery and stakeholder meetings; review their writing; grow scope gradually."

Assumptions & trade-offs: autonomy vs support.

Common mistakes: doing their work for them.

60-second answer: ownership + feedback + pairing.

LE4 · Hard — How do you make a high-stakes, irreversible decision?

What it tests: decision quality.

Approach: classify reversibility → diverse input → scenarios → pre-mortem → criteria → decide → communicate → checkpoints.

Model answer: "Separate reversible from irreversible decisions; gather diverse input; quantify scenarios; run a pre-mortem; set decision criteria in advance; decide; communicate the rationale; define checkpoints."

Assumptions & trade-offs: speed vs certainty.

Common mistakes: treating every decision as irreversible.

60-second answer: reversibility → pre-mortem → criteria → decide.

LE5 · Medium — Your team is burnt out before a critical launch.

What it tests: people leadership.

Approach: protect people → trade-offs → honesty upward → learn.

Model answer: "Cut scope, adjust the date if possible, remove distractions, rotate on-call, be honest with leadership about trade-offs, and run a planning retro afterwards."

Assumptions & trade-offs: commitments to customers.

Common mistakes: pushing harder.

60-second answer: reduce load, communicate, learn.

LE6 · Hard — What would you do in your first 90 days as a PM here?

What it tests: onboarding plan and judgement.

Approach: 30 learn → 60 focus → 90 deliver and plan.

Model answer: “Days 1–30: learn users, data, team, strategy and stakeholders. Days 31–60: identify the top problem and a quick win; build relationships and rituals. Days 61–90: propose an outcome-based plan, deliver one improvement, share learnings.”

Assumptions & trade-offs: urgent issues may compress learning.

Common mistakes: big changes before understanding context.

60-second answer: learn → focus → deliver.